L-C-001081B(GSA-FSS)
April 25, 1972
SUPERSEDING
Int. Fed. Spec. L-C-001081A(GSA-FSS)
November 15, 1967

INTERIM FEDERAL SPECIFICATION

COVER, PILLOW (PLASTIC)

This Interim Federal Specification was developed by the Federal Supply Service, General Services Administration, Washington, D. C. 20406, based upon currently available technical information. It is recommended that Federal agencies use in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE

1.1 Scope. This specification covers the requirements for a plastic pillow cover.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

L-P-375 - Plastic Film, Flexible Vinyl Chloride.
V-F-106 - Fasteners; Slide, Interlocking.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Box, Fiberboard.
PPP-B-665 - Boxes; Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676 - Boxes, Setup.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 191 - Textile Test Methods.
Fed. Test Method Std. No. 406 - Plastics: Methods of Testing.
Fed. Std. No. 751 - Stitches, Seams and Stitchings.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from the established distribution points in their agencies.)

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Standard Test Methods for Vinyl and Pyroxylin Coated Fabrics and Vinyl Sheeting.

(Application for copies should be addressed to the Vinyl Fabrics Institute, 60 E. 42nd Street, New York, New York 10017.)

American Society for Testing and Materials (ASTM) Standards:

D 618 - Conditioning Plastics and Electrical Insulating Materials for Testing.
D 689 - Internal Tearing Resistance of Paper.
D 1239 Resistance of Plastic Films to Extraction by Chemicals.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Incorporated, Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association Inc., Tariff Order Section, 1616 P Street, N.W., Washington, D. C. 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 202, Union Station, 516 W. Jackson Blvd., Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Preproduction sample. When specified (see 6.2), a preproduction sample of the finished pillow cover shall be submitted as directed by the contracting officer for approval.

3.2 Material.

3.2.1 Vinyl film. The material for the pillow cover shall be virgin vinyl film not less than 4 gage (0.004 inch + 10 percent) in thickness before embossing, boilable and contains no ingredients known to be skin sensitizers or irritants and shall conform to the physical requirements as shown in table I, when tested as specified in 4.4. The vinyl film shall meet the resistance requirements specified in 3.2.2 and 3.2.3. The material shall be taffeta embossed on one (top) side.

TABLE I. Physical requirements 1/

Characteristic	Requirement	
Tensile strength, pounds/sq. inch, minimum	2,000	2/
Elongation, percent, minimum	180	
Tear strength, grams/mil., minimum	100	2/
Dimensional stability, percent change, maximum (plus or minus)	7.0	
Rate of burn inches/second, maximum	1.2	
Volatility, percent loss, maximum	2.0	
Soap extraction percent loss 70°C maximum	6.0	
90°C maximum	7.0	

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1/ Tests in table I shall be conducted on the vinyl film after embossing, unless otherwise specified.
2/ Thickness used in calculation shall be the thickness of vinyl film before embossing.

3.2.2 Resistance to boiling water. The vinyl film shall not become brittle or sticky when tested as specified in 4.5.1.

3.2.3 Stain resistance. The vinyl film shall show no effects, other than slight discoloration, to blood, saliva, oil, grease, alcohol or perspiration, when tested as specified in 4.5.4.

3.2.4 Slide fastener. The slide fastener chain shall be made of aluminum alloy having a short pull tab. Material content of components shall conform to that specified in V-F-106. The slide fastener shall have a tape of sufficient width and strength for efficient operation of the fastener and secure attachment to the cover.

3.2.5 Thread. The thread for sewing the slide fastener and seaming shall be white or a clear nylon of a sufficient strength to produce secure sewing.

3.3 Cover construction. At the option of the contractor, the method of construction shall conform to Method A or B.

3.3.1 Method A. The cover shall be fabricated from one panel of the material specified in 3.2.1. The vinyl film shall be folded to form a flat bag. The pillow cover shall be bound on two sides and open end with the material specified in 3.2.1. The binding tape for the sides and open end shall be 1 (+ 1/8) inches wide and shall have a 3/16 (+ 1/16) inch turn-under. One row of stitching shall be used to join the two plies of material and binding together. The slide fastener shall be not less than 19 inches long and shall have a stop on each end. The fastener shall be centered crosswise and shall be securely attached to one end (open end) with binding tape. The slide fastener tape shall be extended 3/4 inches on each end, past the end of the fastener chain. Edges of the slide fastener tape shall be securely sewed on ends and sides, in order to remain fast and secure under normal opening and closing of the fastener.

3.3.2 Method B. The cover shall be fabricated from one panel of the material specified in 3.2.1. The vinyl film shall be folded at one end, heat sealed on the two sides and the open end shall accommodate the slide fastener. Folded end need not be seat sealed. Or as an alternate method of construction, two panels of the material specified in 3.2.1 may be superimposed and one end and two sides heat sealed. The open end shall accommodate the slide fastener. The side hems and end hem (when using two panels) shall be seamed with a minimum of 1/8 inch electronic heat seal on the inside of the cover. The slide fastener shall be centered crosswise on the pillow cover width and the fastener tape shall be securely attached on the inside of the cover by means of an electronic heat seal. The heat seal shall be 1/4 + 1/32 inch in width and shall extend beyond the slide fastener chain for a distance of not less than 1/2 inch at both ends. The slide fastener shall be not less than 19 inches long and shall have a stop on each end. The heat sealed seams shall be neatly made, uniformly heat sealed, continuous and shall not weaken the basic material. Edges of the slide fastener tape shall be securely heat sealed on sides and ends in order to remain fast and secure under normal opening and closing of the fastener. When tested as specified in 4.6, the strength of the bond of the heat sealed seams shall be not less than 65 percent of the breaking strength of the vinyl film.

3.4 Stitches, seams and stitching. All stitches, sewed seams and stitching shall conform to the requirements of Fed. Std. No. 751. All sewing shall be done using thread specified in 3.4.1. The seam type shall be BSc-1 and stitch type 301 of Fed. Std. No. 751. The stitching shall be not less than 1/4 inch from the edge of the binding. There shall be not less than 6 and not more than 8 stitches per inch.

3.5 Size. The pillow covers shall be 20 inches wide by 27 inches long. The following tolerances will be permitted: Width ± 1/4 inch; length ± 1/4 inch.

3.6 Color and finish. The pillow cover shall be white and the embossed side of the vinyl film shall be on the outside of the cover.

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3.7 Workmanship. The finished pillow cover shall be free from defects which detract from its appearance or may impair its serviceability. The cover shall conform to the quality and grade of product established by this specification. Occurrence of defects shall not exceed the applicable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection of components and materials. In accordance with 4.1, the supplier is responsible for insuring that components and materials used were manufactured, examined, and tested in accordance with this specification and referenced subsidiary specification and standards. Thickness test of vinyl film before embossing, as specified in 3.2.1 shall be conducted in accordance with standard commercial practice. Acceptance of this characteristic may be based on contractor's certificate of compliance.

4.3 Inspection for end item lot acceptance. Sampling for inspection and lot acceptance shall be determined in accordance with MIL-STD-105, except where otherwise indicated.

4.3.1 Examination of the end item. The covers shall be examined for visual and dimensional defects.

4.3.1.1 Visual examination of the end item. Defects found during this examination shall be classified in accordance with the following list. The lot size shall be expressed in terms of one pillow cover. The unit of product shall be expressed in the same terms as the lot size. The inspection level for the pillow cover shall be level I and the acceptable quality level (AQL) shall be 2.5 major defects per hundred units and 10.0 total (major and minor combined) defects per hundred units.

TABLE II. Classification of defects

Examine	Defect	Classification Major	Classification Minor
Material	Not as specified.	X	
	Bubble, sharp crease, wrinkle, blister gouge, or other imperfections.		X
	Cut, tear, hole, break, or crack.	X	
	Color not as specified.	X	
	Vinyl film not taffeta embossed on outside of cover.		X
	Objectionable odor (other than odor typical to vinyl chloride).		X
	Patch, splice or repair of plastic material.	X	
	Any spot, stain or other discoloration affecting appearance.		X

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TABLE II. Classification of defects

Examine	Defect	Classification Major	Classification Minor
Slide fastener	Material other than specified.	X	
	Size not as specified.	X	
	Does not function properly or does not operate freely, smoothly, or positively, i.e., binds, catches, or sticks at any point.	X	
	Any scoop which is improperly mated or does not mesh with its corresponding scoops when chain is in closed position.	X	
	Slider goes beyond stop.	X	
	Slider pull becomes detached during operation.	X	
	Slider disengages from either row of scoops when chain is in closed position.	X	
	Ends and sides of slide fastener tape not sewed securely. (Note: Slide fastener shall be opened and closed several times with reasonable pull and tension to insure secure sewing of slide fastener.)		X
Thread	Not as specified.	X	
	Color not as specified.	X	
Construction Method A	Not a single panel of material.	X	
	Binding tape not of the same material as the cover.	X	
Construction Method B	Not a single panel of material.	X	
Heat seal	Open seam of heat seal.	X	
	Poor heat seal resulting in inadequate fusion.		X
	Heat seal seam puckered or distorted in heat sealing.		X
	Heat seal seam misplaced, out of alignment or not on a straight plane by more than 1/4 inch.		X
	Slide fastener tape not heat sealed on inside of cover or reversed, i.e., pull is on the inside of cover.	X	
Stitches, seams and stitching	Not stitch type 301.	X	
	Two or more stitches per inch less or more than required.		X
	Not seam type BSa-1.	X	
	Any open seam.	X	

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TABLE II. Classification of defects (Con't)

Examine	Defect	Classification Major	Minor
Stitches, seams and stitching (Contd)	Loose tension resulting in a loose seam.		X
	Tight tension (stitches break when normal pull is applied).	X	
	Stitching irregular or uneven, resulting in a badly twisted, pleated or bulging edges.		X

4.3.1.2 Dimensional examination. Any measurement deviating from the specified dimension by more than the tolerance shall be considered a defect. The lot size shall be expressed in units of one pillow cover. The sample unit shall be one pillow cover. The AQL for this examination shall be 2.5 defects per hundred units. The sample size shall be selected in accordance with inspection level S-3 of MIL-STD-105.

4.4 Inspection of preparation for delivery requirements. An examination shall be made to determine that applicable packaging, packing, and marking requirements are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 4.0 defects per hundred units.

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence or method of application.
Materials	Any component missing. Any component damaged, affecting serviceability.
Workmanship	Inadequate application of components, such as incomplete closure of case liners, container flaps, loose strapping, inadequate stapling. Bulging or distortion of containers.

4.5 Tests. The methods of testing specified in Fed. Std. No. 406, table IV, and paragraphs 4.5.1, 4.5.2, 4.5.3, 4.5.4, 4.6, wherever applicable shall be followed. The physical and chemical values specified in section 3 apply to the average of the determination made on a sample unit for test purposes as specified in the applicable test method. The sample unit shall be one pillow cover. The lot size shall be expressed in units of one pillow cover each. The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified. The sample size (number of sample units) shall be in accordance with Table III. Acceptance of the characteristics listed in table IV, as applicable, may be based on component testing with a contractor's certification of compliance accompanied by actual test, inspection or other verifiable data in lieu of end item testing.

TABLE III

Lot size	Sample size
800 or less	2
801 thru 22,000	3
22,001 and up	5

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TABLE IV. Tests 1/

Characteristic	Requirement reference	Test Method
Tensile strength	Table I	Method B - 1031 2/
Elongation	Table I	Method B - 1013 2/
Tear strength (Elmendorf Method)	Table I	ASTM D689
Dimensional stability (Dimensional change at elevated temperatures)	Table I	Paragraph 4.5.3
Rate of burn	Table I	Paragraph 4.5.2
Soap extraction	Table I	ASTM D1239
Resistance to boiling water	3.2.2	Paragraph 4.5.1
Volatility	Table I	6081 2/
Stain resistance	3.2.3	Paragraph 4.5.4
Bond strength of heat sealed seams	3.3.2	Paragraph 4.6

1/ Acceptance of the characteristics listed in Table IV may be based on component testing with a contractor's certification of compliance accompanied by actual test, inspection or other verifiable data.

2/ Fed. Test Method Std. No. 406.

4.5.1 Resistance to boiling water. The vinyl film shall be placed in boiling water for 60 minutes.

4.5.2 Rate of burn. A standard Flame Tester, natural or butane gas, and heavy duty cotton thread are required for this test. Five specimens from each direction 3 inches wide by 9 inches long, free from wrinkles and/or folds are to be tested. Conditioning to conform to procedure B of ASTM-D-618-61, Methods of Conditioning Plastics and Electrical Insulating Materials for Testing. Thread the holder to depress switch actuators at least 1/4 inch from edge of holder. Thread each actuator separately, thread passing down through J-slots and under upper jaws, so that the thread is adjacent to specimen when holder is closed. Insert into holder so it extends down into lock springs and is held firmly between two wires at open end of burning channel. Wires insure that end of specimen is correct distance from igniter flame. Prior to introducing specimen and holder into cabinet, set switches for automatic timing to provide 1/2 inch flame. (Specimen is in place when surface is 9/16 inch from tip of needle and the flame is barely flattened against specimen.) With hood fan off, clocks zeroed and door closed, insert specimen holder at a constant rate, takes about 1/2 second to travel down rails by gravity. Read burning time from stop clock and calculate rate of burn as follows:

$$\frac{6}{\text{Seconds of burning}} = \text{Inches/Second}$$

Reject results that deviate from mean value of all tests if deviation of doubtful value is more than 5 times the average deviation from the mean by excluding the doubtful value. Discard doubtful values and retest.

Samples which ignite but extinguish are to be reported as IBE, otherwise report rate of burn in inches per second.

4.5.3 Dimensional stability (dimensional change at elevated temperatures). Dimensional stability shall be determined by exposing a 10-inch square sample of film to 212° + 2°F. (100° + 1.1°C) for 30 minutes + 1 minute in a circulating air oven. The sample shall be cut from the center of the film sheet with a template accurate to + 0.02 inch in all directions. The sample shall be placed between sheets of heavy wrapping paper, 15 inches square, lightly dusted with talc. The paper sheets shall be clipped or stapled together, care being taken not to hold or restrict the test sample. The sandwich shall then be placed horizontally in an oven. After the oven exposure, the sample shall be cooled to room temperature and measured to the nearest 0.01 inch along both central axes.

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4.5.4 Stain resistance.

4.5.4.1 Stain resistance to blood. Pour 5 milliliters (ml.) of blood (human, warm-blood animal, or fowl) on a specimen of material 12 inches square. Spread the blood to a thin layer. After drying, the blood shall be allowed to remain in contact with the plastic film for 24 hours. The specimen shall then be washed off with a cloth or sponge using soap and water. Observe for stains or discoloration.

4.5.4.2 Stain resistance to saliva. Use the same method as specified in 4.5.4.1 except that human saliva shall be substituted for blood.

4.5.4.3 Stain resistance to oil. Pour 5 ml, of light mineral oil, U.S.P. or castor oil, U.S.P. on a specimen of material 12 inches square. Spread the oil to a thin even layer. Allow the oil to remain in contact with the plastic film for 24 hours. Wash off with a cloth or sponge using soap and water. Observe for stains or discoloration.

4.5.4.4 Stain resistance to grease. Use the same method as specified in 4.5.4.3, except that 5 grams of petrolatum shall be substituted for the oil.

4.5.4.5 Stain resistance to alcohol. Pour 5 ml. of rubbing isopropyl alcohol or 70 percent ethyl alcohol on a specimen of material 12 inches square. Spread to a thin layer. Permit the alcohol to evaporate at room temperature. Repeat the procedure every half hour for 4 hours. After 4 hours, wash with a cloth or sponge using soap and water. Observe for stains or discoloration.

4.5.4.6 Stain resistance to perspiration. Pour 5 ml. of acid solution as specified in method 5680 of Fed. Test Method Std. No. 191 on a specimen of material 12 inches square. Spread the solution to a thin layer. After drying, allow the residue to remain in contact with the plastic film for 24 hours. Wash off with a cloth or sponge using soap and water. Observe for stains or discoloration. Repeat the test using alkaline solution as specified in method 5680 of Fed. Test Method Std. No. 191.

4.6 Bond strength. The bond strength of the heat sealed seams shall be tested as specified in Federal Specification L-P-375, except that tests shall be made on one side heat sealed seam and, when applicable, the end heat sealed seam.

(a) Two specimens, 1 inch wide by 7 inches long, shall be cut from the side heat sealed seam.

(b) One specimen, 1 inch wide by 7 inches long, shall be cut from the end heat sealed seam, when applicable.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Each cover, folded in accordance with commercial practice, shall be inserted in a polyethylene bag and the bag sealed.

5.1.1.1 Intermediate packaging. Twelve covers shall be intermediate packaged in a paperboard box conforming to PPP-B-566, PPP-B-665 or PPP-B-676.

5.1.2 Level C. The covers shall be packaged in accordance with the supplier's commercial practice.

5.2 Packing. Packing shall be level A, B, or C as specified (6.2).

5.2.1 Level A. Twelve (12) intermediate packages of covers shall be packed in a close-fitting box conforming to PPP-B-636, class weather-resistant. The box shall be closed in accordance with the appendix to the box specification.

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5.2.2 Level B. Twelve (12) intermediate packages of covers shall be packed in a close-fitting box conforming to PPP-B-636, class domestic. The box shall be closed in accordance with the appendix to the box specification.

5.3 Marking.

5.3.1 Civil agencies. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military activities. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The pillow cover covered by this specification is intended primarily for use in hospitals for the protection of pillows from soilage.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) When preproduction sample is required (3.1).
(c) Selection of applicable level of packaging and packing required (5.1 and 5.2).

6.3 Safety hazards for hospital use. Because of possible electrostatic buildup the plastic pillow covers, covered by this specification should not be used when oxygen is administered by the tent method or where flammable anesthetics are used (National Fire Protection Association No. 56 - Code for the Use of Flammable Anesthetics (current edition). Local safety regulations should be followed regarding materials to be used under these conditions.

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